

TOI-4087b

R-2051

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_260225 · created 2026-08-09T00:29:25+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461097.0245 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1344 +/- 0.005
Transit depth (Rp/Rs)^2	1.81 +/- 0.13 [%]
Orbital Inclination (inc)	89.52 +/- 1.02
Airmass coefficient 1 (a1)	1.249 +/- 0.015
Airmass coefficient 2 (a2)	-0.14 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	#3 - [515, 164]
Optimal Aperture	3.58
Optimal Annulus	9.08
Transit Duration (day)	0.0878 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1344 ± 0.005 vs 0.10754068997360952 (deviation 2.9σ)
Duration fit vs published	0.0878 d vs 0.1242 d (deviation 6.14σ)
Mid-time O–C	-18.6 min
Depth SNR	14.5
Residual scatter	1.17 %

Light curve

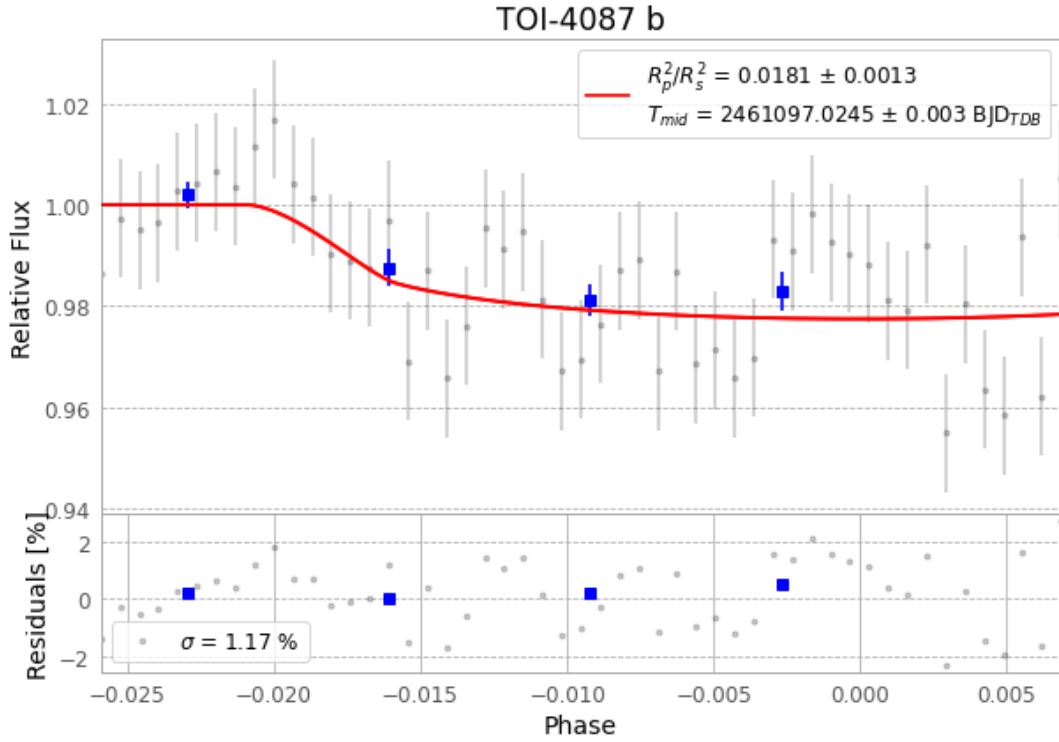


Figure 1 — FinalLightCurve_TOI-4087_b_2026-02-25.png (SHA-256 0970e899f665fd20...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 64], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c430fb84fcc05e4e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

52 FITS frames (34 MB), TOI4087260225103015.FITS → TOI4087260225130315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-260225.log (399bd68e4ba7...), FinalLightCurve_TOI-4087_b_2026-02-25.png (0970e899f665...), FinalLightCurve_TOI-4087_b_2026-02-25.csv (421c3756a749...)

Compute resources

Wall time 146.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:29Z.